

Weeks 2-10

Grades 4-5

Investigate all cases, prove an obstruction, or explain an optimum.

Grade bands are entry points. Begin with objects, then try to explain what happens.

Give one page at a time. The packet is a menu, not a checklist for one hour.

Keep facilitator solutions separate from student pages.

Week and investigation	PDF pages
02 Switches and lamps	2-4
03 Repeating secret machines	5-7
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Print US Letter, single-sided, at 100% / Actual Size.

Page ranges above count this cover as page 1. Bookmarks open each week.

Weekly footers retain their own page numbers and activity IDs for the use log.

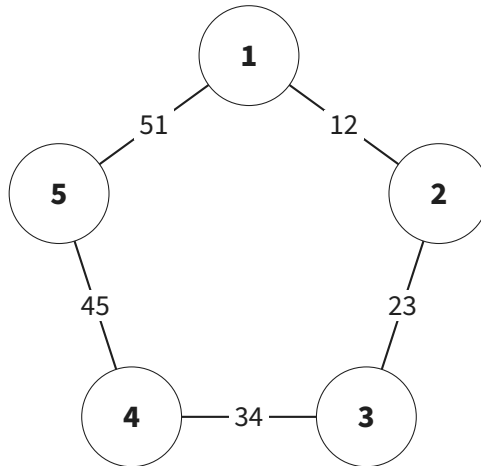
Revised September 20, 2026. Planned activities; record actual use after teaching.

The cheapest light show

Name: _____

Date: _____

All lamps begin OFF. Press an edge to flip its two end lamps. Pressing 12 flips lamps 1 and 2. Minimize the **number of presses**.



1. Find two ways.

Light **only lamps 1 and 3**. Find a shortest solution and a different solution. Reset before each attempt.

Two move lists; why the shorter one is best:

2. Four lamps.

Now light **1, 2, 3, 4** and leave 5 OFF. Find two ways again. Can you prove your best way is best?

My best construction and a lower bound:

Try it with objects. Explain by pointing, talking, or drawing.

Are there other solutions?

Pressing the same edge twice cancels. The order of presses does not matter: check this at each lamp. So record just the edges pressed an **odd** number of times.

3. Start with a choice.

Target: only 1 and 3 ON. Decide whether to use edge 51. Then use lamp 1’s target to decide 12, lamp 2’s target to decide 23, and so on. Why is each later edge forced, with no new choice? Does the last lamp check work? Explain why the two first choices cover every reduced solution.

First choice	51	12	23	34	45	Check lamp 5
Do not use 51	no					
Use 51	yes					

4. The missing-edges trick.

Take one solution. Press exactly the edges it did **not** use. Does that give the same target? Explain at a lamp, which touches two edges.

Why the trick works, and why there cannot be a third edge set:

Try to break your rule: Repeat the table for a target with only lamp 1 ON. Where does the consistency check fail?

Try it with objects. Explain by pointing, talking, or drawing.

A theorem for every target

5. Possible or impossible?

Conjecture: a target is reachable exactly when it has an even number of ON lamps. Prove the impossible direction. For the possible direction, pair the ON lamps and join each pair by a path.

Why paths give every even target:

6. A surprising speed limit.

On this five-lamp ring, can every reachable target be made in **at most two presses**? Use the two complementary edge sets from page 2.

A construction or a reason that covers all targets:

7. Change the ring.

Draw a six-lamp ring. Light only opposite lamps 1 and 4. What is the fewest presses? Can the two complementary solutions tie?

Six-lamp experiment and explanation:

Try it with objects. Explain by pointing, talking, or drawing.

How long until every letter returns?

Name: _____

Date: _____

A substitution key uses A, B, C, D, E once each as inputs and once each as outputs. Apply the key to every letter **simultaneously**. Repeat the same key each turn.

Input	A	B	C	D	E
Output	B	C	A	E	D

1. Test this machine.

Start with ABCDE. Record successive messages until the **entire** message first returns. Draw the arrow loops separately.

Turn	Message	Turn	Message
0	ABCDE	4	
1		5	
2		6	
3		7	

The loops, and why their clocks must agree:

2. Make it slower.

Can a different five-letter key take longer to return? A five-letter loop is one candidate. Try several keys; keep a record of the loop sizes.

My best key, its loops, and its return time:

Try it with objects. Explain by pointing, talking, or drawing.

Prove the best is best

3. Replace a huge list by a small one.

Cover the table first. Arrange your five letter cards into loops, working by largest loop: 5, then 4, then 3, then 2, then 1. List how the remaining cards can split without a larger loop. Uncover and compare. A fixed letter is a loop of size 1; rearranging sizes gives no new case.

Loop sizes	First common return time
5	
4 + 1	
3 + 2	
3 + 1 + 1	
2 + 2 + 1	
2 + 1 + 1 + 1	
1 + 1 + 1 + 1 + 1	

4. Explain the classification.

Why does following arrows eventually bring a letter back to **itself**, rather than merge into another loop? Where does using each output once matter? Why does your largest-loop list omit no possible split?

Why all keys are covered, and the largest return time:

A useful language: A loop of size 3 returns on turns 3, 6, 9, The whole key returns at the first turn that is a multiple of *every* loop size. Mathematicians call this the least common multiple.

Try it with objects. Explain by pointing, talking, or drawing.

Does the order of two keys matter?

Key P swaps A and B and leaves C, D, E alone. Key Q swaps B and C and leaves A, D, E alone. “P then Q” means apply P first, then Q to its output.

5. Compare two pipelines.

Start with ABCDE each time. Try P then Q. Reset and try Q then P. Do they agree? One differing letter is enough to disprove agreement.

	After first key	After second key
P then Q		
Q then P		

An arrow picture explaining the difference:

6. Repeat the entire pipeline.

P alone returns in two turns; so does Q. If a turn is now **P then Q**, how many turns until everything returns? Find its loops.

My answer and a loop proof:

Go further: Make two different nonidentity keys whose order *does not* matter. What if they move completely separate sets of letters? Can you undo P then Q using the same two keys?

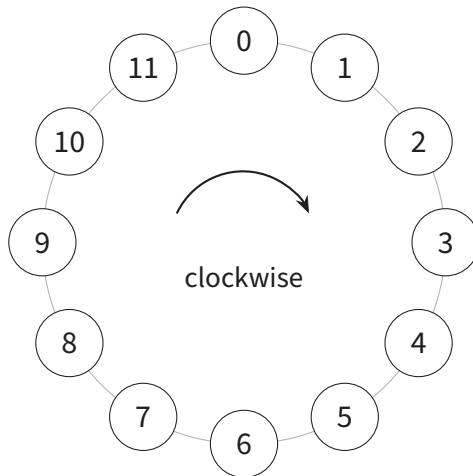
Try it with objects. Explain by pointing, talking, or drawing.

The twelve-place code

Name: _____

Date: _____

Use positions 0 through 11 clockwise. A shift $+k$ moves every input k places clockwise, wrapping after 11. Decoding reverses that shift. A turn repeats the **same** shift.



1. A broken message?

Input 9 became output 1. Find the shift $+k$, where k is between 0 and 11. Check whether input 2 could become 6 under the same shift. Decode the output sequence **1, 4, 10**.

My key, check, and decoded message:

2. Follow one input.

Starting at 0, repeat that shift until the first return. Then try starting at 1. Find **all** its separate loops, including every position exactly once.

All loops, and the first return time:

Try it with objects. Explain by pointing, talking, or drawing.

Classify every shift

3. Find a rule, not just a table.

Try enough shifts to spot a rule: start at 0, find the first positive return, and count separate loops. Use clockwise/counterclockwise symmetry. The full table is optional; move to page 3 when ready to explain.

Shift	Turns to return	Number of loops	Visits all 12?
0			
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			

4. Predict before you test.

On an 18-place ring, predict the return times for +6 and +5. What number connects the ring size, the step size, and the number of loops?

My conjecture and evidence:

Try it with objects. Explain by pointing, talking, or drawing.

Why the rule works

5. Count distance around the ring.

After t turns of $+k$ on an n -place ring, you have moved kt spaces. You are back at the start exactly when kt is a whole multiple of n . Use this to explain $+4$ on a 12-ring.

The first return and why none earlier works:

6. Shrink the common factor.

For $+8$ on 12 places, return means 12 divides $8t$, or 3 divides $2t$. Draw three equal t -strips: together they total $3t$. If two strips total a multiple of 3, why does removing them leave t a multiple of 3? Explain the first return.

My strip argument; optional general rule:

7. Undoing is different from visiting.

The $+4$ rule does not visit every place from 0. Does it still give a reversible code on *all* twelve inputs? Find its inverse. Then compare $+4$ followed by $+7$ with $+7$ followed by $+4$. Explain for every starting place.

Inverse, combined shift, and why the order agrees:

Optional, with an adult: Use the guide's subtraction-of-strips argument to justify coprime cancellation, then prove the gcd rule. Until then, label the general rule a conjecture or a supplied theorem.

Try it with objects. Explain by pointing, talking, or drawing.

A city with a certificate

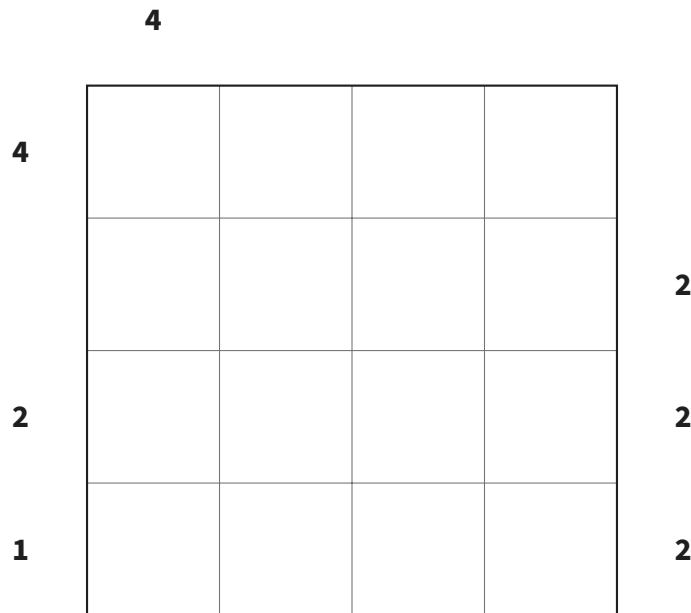
Name: _____

Date: _____

Use heights 1–4 once in each row and column. Outside numbers count visible towers from that end.
Taller towers hide shorter towers behind them.

1. Solve with towers.

Build any legal city first. Then solve this puzzle. Record a forced step and explain it to a partner.



My forced step and the clue that forces it:

2. Two clues disappear.

Cover the top 4 and the left 4. Is the city still the only answer? A failed search is not yet a proof. Save your city; continue on the next page.

Build first. Explain by pointing, talking, or drawing.

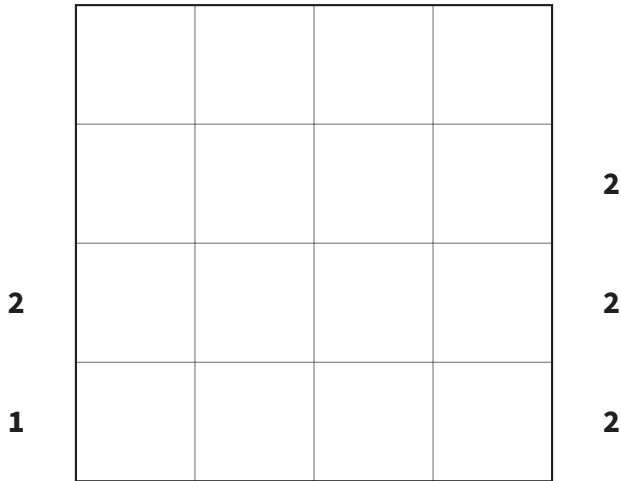
Can five clues force everything?

Name: _____

Date: _____

3. Rebuild using only these clues.

There are no top or bottom clues. Work row by row; make a list of possible rows when a step is uncertain.



A route into a proof: The bottom row begins with 4. Looking from the right, exactly one tower is seen before 4. Where must height 3 be? Which two bottom rows remain?

Record both possible bottom rows; keep both for now.

Next: List row 3’s possibilities using both its clues. Pair each with a bottom row, rejecting column repeats. For each surviving pair, test row 2’s right clue. Only now can you rule out a bottom row.

Bottom row	Row 3	Possible row 2 (or none)

Finish the proof: How is the top row forced? Explain why your cases cover every city.

Build first. Explain by pointing, talking, or drawing.

Necessary is not the same as smallest

Name: _____

Date: _____

Optional continuation: A complete uniqueness proof on page 2 is a good stopping point. Continue with an adult when ready.

4. Check five certificates.

First try deleting the left clue on row 3 yourself. Then check the alternatives below. L3 means the left clue on row 3; R2 means the right clue on row 2. Each slash starts the next row.

Delete	Alternative city, top row first	Checked
L3	1234 / 3412 / 2341 / 4123	
L4	1234 / 4123 / 3412 / 2341	
R2	1324 / 2431 / 3142 / 4213	
R3	1324 / 3142 / 2431 / 4213	
R4	1234 / 2143 / 3412 / 4321	

For each city, check every row and column, all four retained clues, and that it differs from your target. Restore the deleted clue before the next check. Why does this prove *each* of the five clues is necessary in this set?

A clue set is **minimal** if it forces one city and deleting any clue destroys uniqueness. A **minimum** clue set uses the fewest clues among perimeter-clue sets for the same target city.

5. A different set uses only three clues.

Remove every old clue. Use only left clues **4, 3, 2 on rows 1, 2, 3**. Build the city. Row 1 is forced; list row 2's possibilities using its clue and the columns, then do row 3. Why is row 4 forced?

Why these three clues force the same city, and what that says about five:

Our conclusion: Three works, so five is not the smallest. Have we proved that two can never work? That is a separate question; the facilitator has a computer check for this target.

Build first. Explain by pointing, talking, or drawing.

Build a guaranteed code breaker

Name: _____

Date: _____

Two colors: R and B. Repeats allowed. A test earns one point for each color in the correct position. You hear **only the total**, not which positions match.

1. Play four places.

Try a hidden code. Then build a method that always identifies it in at most **four scored tests**. You may name the answer without paying for a final test.

Test	My code	Score	Still possible
1			
2			
3			
4			

2. Explain a baseline method.

Test RRRR. Then separately change place 1, place 2, and place 3 to B, keeping every other place R. Explain how the fourth place is recovered.

What each score change tells me, and why the method covers every secret:

A real optimization question: Four tests are enough. Could a cleverer, adaptive method always manage three? On the next page, play the role of a difficult secret maker.

Build first. Explain by pointing, talking, or drawing.

The secret maker fights back

Name: _____

Date: _____

Two colors: R and B. Repeats allowed. A test earns one point for each color in the correct position. You hear **only the total**, not which positions match.

3. Make the first answer difficult.

After RRRR, answer 2. These are the six possible secrets. Group them by their score for each proposed second test.

BBRR BRBR BRRB

RBBR RBRB RRBB

Second test	Score groups	Largest group	Its size
BBRR			
BBRR			
BBBR			
BBBB			

4. Only one test remains.

For BRRR, let the answer leave three candidates. Can one last test separate those three? For BBRR, let the answer leave four candidates. Can one last test separate those four?

Try final tests; explain why each difficult branch defeats them.

Three-secret hint: Use BBRR, BRBR, BRRB. Which position is fixed? What is the only feature that changes? A test asks whether the moving B falls in one of its B positions. How many scores can that give?

Four-secret hint: All six original candidates contain exactly two Bs. How many scores can one test produce on them? Is that enough to separate four candidates?

Build first. Explain by pointing, talking, or drawing.

Turn the experiment into a proof

Name: _____

Date: _____

5. Cover every possible second test.

Second tests have 0, 1, 2, 3, or 4 Bs. Changing position names does not change the problem. Swapping every R and B in a test replaces its score by $4 - \text{score}$. Use those facts to explain why the cases on page 2 cover everything.

Why one or three Bs leaves a hard three-secret branch:

Why two Bs leaves a hard four-secret branch:

6. What about a different first test?

Rename the two colors separately at selected positions, in both every secret and every test. Can you turn any first test into RRRR without changing which places match?

My conclusion: the smallest guaranteed number of tests, and why:

A short method proves an **upper bound**. A difficult branch that defeats every shorter method proves a **lower bound**. When they meet, we know the optimum.

Go further: Must five places need five tests? The extra page contains a surprise.

Build first. Explain by pointing, talking, or drawing.

A pattern is a conjecture

Name: _____

Date: _____

Two players alternate. Take **1, 3, or 4** counters each turn. Taking 2 is illegal. Never take more than remain. Taking the last counter wins.

1. Build the evidence.

Play from 10 and 14. Then label positions W or L from the viewpoint of the player about to move. Put L at zero.

0	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27

2. Make every label earn its place.

For W, give one legal move to L. For L, check all legal moves go to W. Find a disputed position and defend it.

What repeat do you conjecture? State it exactly.

Next question: Is seeing four repeats a proof that the pattern continues forever? What feature of the rules could force it?

Build first. Explain by pointing, talking, or drawing.

Prove the repeat forever

Name: _____

Date: _____

Two players alternate. Take **1, 3, or 4** counters each turn. Taking 2 is illegal. Never take more than remain. Taking the last counter wins.

3. Two obligations.

Your proposed losing piles must satisfy **both** tests: (a) no legal move goes from one proposed L to another; (b) every other positive pile has a legal move to a proposed L.

4. Use groups of seven.

Write a pile as some whole groups of seven plus a remainder from 0 to 6. Complete the table using your conjecture.

Remainder	W or L	Take this many	New remainder
0			
1			
2			
3			
4			
5			
6			

Why do the two obligations prove a winning strategy?

Check the boundary: Your suggested move must be legal even for the smallest positive pile in that remainder class. Why does a decreasing pile eventually reach zero?

Build first. Explain by pointing, talking, or drawing.

Change a rule; challenge a proof

Name: _____

Date: _____

New game: take **1, 3, or 5**. Two players alternate. Taking the last counter wins. This is not “take any number from 1 to 5.”

5. Predict before playing.

Who can force a win from 6? From 7? From 100? Test small piles, then find one explanation for every pile.

My new losing piles and both parts of my proof:

6. How much memory does the old game need?

Return to moves 1, 3, 4. Copy a stretch of your labels onto a straight strip. Put a paper frame around **four consecutive W/L labels**. Which three cells decide the next label? Add that label, then slide the frame one cell right. Find a matching window elsewhere and show why both next labels must agree.

Why identical four-label windows have identical futures:

Go further: A four-label window has at most $2^4 = 16$ patterns. Explain why some window must recur. Does this prove repetition starts at zero, or only that it eventually starts?

Build first. Explain by pointing, talking, or drawing.

Three piles break the mirror

Name: _____

Date: _____

Build three piles. On a turn, remove any positive number of counters from exactly ONE pile. Take the last counter to win. Use ordinary counters, not bundles, for actual play.

1. Challenge an adult.

Try piles $(1, 2, 3)$, then $(1, 2, 4)$. Change who moves first. Which starts seem to be traps for the player receiving them?

Record moves, replies, and a conjecture.

2. Compare two kinds of balance.

With only two piles, matching sizes lets you copy your opponent. With three piles, investigate $(1, 1, 1)$, $(1, 1, 2)$, $(1, 2, 3)$, and $(2, 4, 6)$. Is an even *total* enough to make a trap?

Find evidence against a rule that almost works.

3. Trade for bundles.

Use labels 1, 2, and 4. Write each pile as a sum using each label at most once: for example, 5 is $4+1$. For each label, count how many piles use it. Which starts have an even count for every bundle?

Try before turning the page. An adult can make the bundle labels with scraps of paper.

Try it with objects. Explain by pointing, talking, or drawing.

Balance the binary columns

Name: _____

Date: _____

Represent each pile using bundles 4, 2, 1. Write 1 if that bundle is used and 0 if not. Call a position **balanced** when every column contains an even number of 1s. Zero 1s counts as even.

Pile	4	2	1	Pile	4	2	1
1	0	0	1	1	0	0	1
2	0	1	0	2	0	1	0
3	0	1	1	4	1	0	0

4. Repair the unbalanced position.

Change just one pile in (1, 2, 4) to a *smaller* number, making every column even. What move does this mean with counters?

5. Test your rule.

For each start, either give a move to balance or say it is already balanced. Then play against someone trying to break your prediction.

Start	4/2/1 records	Move or already balanced?
(2,4,6)		
(3,4,5)		
(1,5,7)		
(3,5,6)		

Can a legal move from balance leave every column even? Why?

Vocabulary after discovery: The column parities form the *Nim sum*, also called bitwise XOR. A balanced position has Nim sum zero.

Try it with objects. Explain by pointing, talking, or drawing.

Why must the strategy work?

Name: _____

Date: _____

6. The highest wrong column.

Suppose some column is odd. Find the largest bundle whose column is odd. At least one pile uses it. In that pile:

1. Keep all larger bundle choices unchanged.
2. Remove this bundle.
3. Choose the smaller bundles to make *every* column even.

Try this recipe on $(3, 4, 5)$. Then try $(7, 10, 12)$, adding an 8-column.

My moves, and why they fix all the columns:

7. Two missing pieces of a proof.

Why is the new pile *smaller*, even if you add some smaller bundles? Why can the game not continue forever?

A reason that works for any pile sizes:

8. Design a trap.

Choose any first two piles. Can you choose the third to make balance? Could you do it in two different ways? Give your puzzle and its explanation to an adult.

My designed start and why its third pile is forced:

Try it with objects. Explain by pointing, talking, or drawing.

A straight line in many rooms

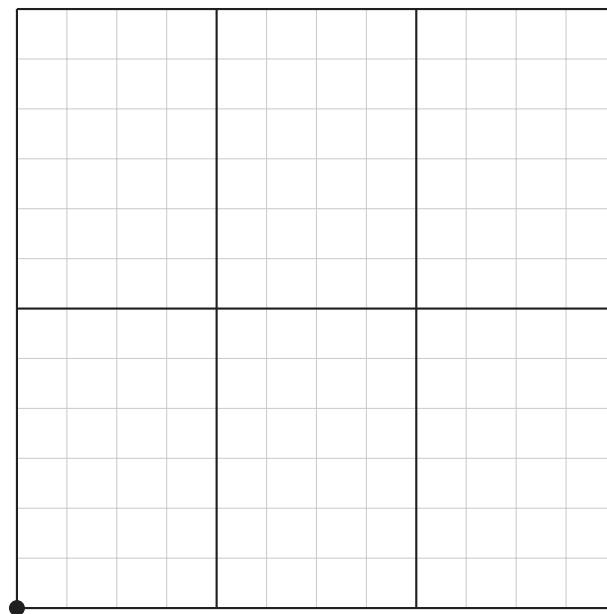
Name: _____

Date: _____

Table sizes are width w by height h . Start bottom-left at 45 degrees. Reflect at walls; stop at the FIRST corner. Count a wall bounce, but not the starting or ending corner.

1. Trace, then unfold.

Find the route on a 4×6 table. If this trace is familiar, begin with the copied rooms and explain your prediction. Mirror copies let a bounce become a straight continuation. On the copied rooms below, draw the diagonal to the first meeting of a vertical and horizontal room wall.



each room is width 4, height 6

2. Name the meeting distance.

Call that horizontal distance L . Why must L be a multiple of *both* w and h ? Why do we take the *least* positive common multiple?

Why a corner means a simultaneous wall meeting:

Try it with objects. Explain by pointing, talking, or drawing.

Predict every ending corner

Name: _____

Date: _____

3. Fold the rooms back.

If the straight line crosses $m = L/w$ whole room widths, where does it end: right or left? If it crosses $n = L/h$ whole room heights, where does it end: top or bottom?

w	h	L	m	n	Corner
4	6				
6	9				
6	10				
8	14				

4. Two even sides: is that enough?

Compare 4×6 with 6×10 . Both have even side lengths. Why can that rule give the wrong answer?

5. A corner too soon.

Suppose m, n shared a divisor $d > 1$. How many whole widths and heights would the path have crossed at distance L/d ? Why would that already be a corner? Use this to rule out a first ending at bottom-left, which needs both counts even.

An earlier corner would mean:

6. Optional: shrink the side lengths.

With an adult, write $w = ga, h = gb$, where g is their greatest common divisor. The coprime rule says: if a, b share no factor greater than 1, b divides at only when it divides t . For $a = 2, b = 3$, explain this using $3t - 2t = t$. Ask for the general argument. Now $L = gat$ must also be a multiple of gb . Why is the least possible t equal to b ? Conclude $L = gab, m = b, n = a$.

The corner rule in reduced side lengths a and b :

Try it with objects. Explain by pointing, talking, or drawing.

How many bounces is best?

Name: _____

Date: _____

7. Count walls instead of drawing.

Before the final corner, how many vertical room walls does the unfolded line cross? How many horizontal room walls? Could a meeting be counted twice before the first corner?

A formula in m,n (or a,b after task 6), with a reason:

8. Optional proof continuation: inverse design.

Design a whole-number rectangle ending at the **top-right after exactly four bounces**. Find all the possible reduced width/height pairs. Then make a second size with the same shape.

All reduced pairs, and why there are no others:

9. An impossible order?

An adult requests a table ending at the top-right after exactly **three** bounces. Find one, or prove it cannot exist.

My construction or obstruction:

Researcher's habit: Distinguish a table you failed to find from a table you can prove does not exist.

Try it with objects. Explain by pointing, talking, or drawing.

When does a whole-map walk exist?

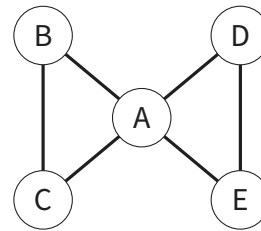
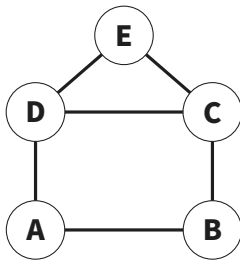
Name: _____

Date: _____

An Euler trail uses every edge exactly once, without jumping. A **closed** Euler trail returns to its start. Vertices may be revisited. Our maps are connected: every edge belongs to the same reachable network.

1. Try the house.

Find an Euler trail on the left map. Count edges at each vertex. Why must its endpoints be C,D? Why can it not close?



2. Insert a loop with your hands.

On the right map, walk A–B–C–A, marking used roads. Another loop remains! Write A–B–C–A on one paper strip and A–D–E–A on another. At a visit to A, insert the second route into the first. Walk your joined route: did it use all six roads once?

My joined route:

Optional proof conversation. On any connected all-even map, follow unused edges until stuck. Why can that happen only at the start? If roads remain, why does some vertex on your closed walk touch one? Start another loop there and insert it. Why must this finish?

Exactly two odd vertices? Add one temporary edge between them. Build a closed route in the all-even map, then cut the route at that edge and remove it. What are the endpoints of the remaining route?

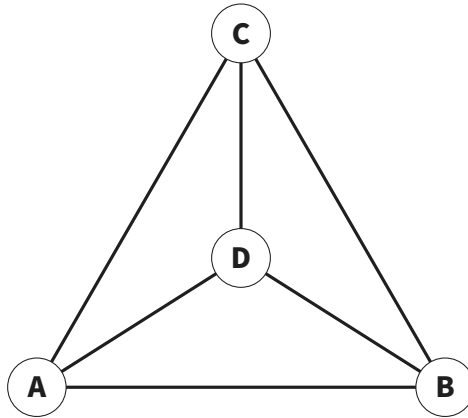
Try it with objects. Explain by pointing, talking, or drawing.

The shortest delivery round

Name: _____

Date: _____

Now deliver along every road **at least once** and return to the start. Reusing roads is allowed. Each road costs 1 step. Minimize total steps. D is a real junction; each drawn segment is a separate road.



3. Find a good route.

Start at A. Record the vertices you visit and count steps. Can you improve your first route?

My best complete route and its length:

4. Prove nobody can beat it.

There are six roads. All four vertices have odd degree. Think of each repeated crossing as adding a second copy of that road. How many odd vertices can one added copy change?

A lower bound that matches my route:

Try it with objects. Explain by pointing, talking, or drawing.

Turn a route into an algorithm

Name: _____

Date: _____

5. Repair the odd vertices in pairs.

List all ways to split A, B, C, D into two pairs. For each pairing, duplicate the road joining each pair. What happens to every degree? Why does a closed Euler trail now exist?

Pairing	Repeated roads	Total length
1		
2		
3		

6. Must the walk be closed?

The mail truck may finish anywhere now. What is the fewest steps needed to visit every road? Build a route and prove its length is best.

My open route and matching lower bound:

7. Become a puzzle designer.

Draw a connected map with **six odd vertices**. Join different dots; parallel roads are allowed. Every road costs 1. Can you design one whose shortest closed delivery round needs exactly three extra steps? Save a route and a lower-bound argument with your puzzle.

A design, or the key idea I will test:

Try it with objects. Explain by pointing, talking, or drawing.